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Exploring the Effectiveness of Institutional Policies and Regulations for Generative AI Usage in Higher Education

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ABSTRACT

As generative artificial intelligence (Gen AI) becomes more integrated into higher education, institutions struggle to communicate clear and effective policies. This multiple-methods study examines survey and interview data collected with 124 undergraduates and seven faculty members from two U.S. R1 universities regarding their perceptions of institutional AI policies, supportive resources and challenges. Students reported that while classroom-level policies are recognised, institutional guidelines remain unclear. Similarly, faculty highlighted the implementation challenges they face due to practical constraints and insufficient training. Additional findings show that students aware of AI policies are less likely to use AI for writing and research; disciplinary and gender differences in institutional resources and policy awareness. Faculty interviews revealed themes such as AI use in research and varied colleague attitudes. Although this study is limited by the number of institutions and sample size, it highlights the need for ongoing research to support policy refinement.

Artificial Intelligence (AI) has significantly evolved over the past few decades, transitioning from traditional AI systems to advanced generative AI (GenAI) models. Traditional AI focused on pattern recognition, prediction and structured decision-making (Johnston 2024). It began with rules-based systems (1950s–1980s) using predefined ‘if-then’ logic, exemplified by systems like MYCIN for medical diagnosis and the General Problem Solver (GPS) which was designed to mimic human problem-solving (Johnston 2024). Machine learning (1990s–2010s) replaced rigid rules with data-driven learning through supervised, unsupervised and reinforcement techniques (Johnston 2024). The deep learning revolution (2010s) introduced neural networks like convolutional neural networks (CNNs) for image recognition in autonomous vehicles and recurrent neural networks (RNNs) for speech processing (Johnston 2024).

Unlike traditional AI's predictive capabilities, GenAI represents a fundamental shift from analysis to content creation to produces original text. This advancement was driven by transformer models, introduced in the ‘Attention is All You Need’ (Vaswani et al. 2017), which use self-attention mechanisms for superior context processing (Johnston 2024). Key tools include large language models (LLMs) like GPT (Generative Pre-trained Transformer) for text generation and DALL-E for image creation (Johnston 2024). This creative capacity distinguishes generative AI from its analytical predecessors (Johnston 2024), and has fundamentally transformed education and raised new policy challenges.

GenAI use among students is rapidly increasing globally; a 2025 survey in the U.K. reported that 92% of students use tools like ChatGPT. However, only one-third have received formal

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training, raising concerns about inadequate institutional guidance (The Guardian 2025). Similarly, a study involving 457 students and 180 faculty members across multiple Hong Kong institutions revealed overall positive attitudes toward GenAI, but also concerns about academic integrity, ethical implications and the need for AI literacy training (Chan 2023). While many institutions have begun to establish policies, students often remain unsure whether or when they are allowed to use GenAI in coursework due to unclear or inconsistently communicated regulations (Mowreader 2024).

1 | International Policy

The emergence of generative artificial intelligence (GenAI) has prompted diverse and uneven policy responses worldwide. In May 2023, UNESCO surveyed over 450 institutions across its global networks of Associated Schools and University Chairs. Institutions from Europe and North America accounted for 44% of responses, followed by Asia and the Pacific (23%), Latin America and the Caribbean (17%), Africa (11%) and the Arab States (5%). The results revealed that fewer than 10% of surveyed institutions had formal policies or guidance on AI use in education (UNESCO 2023). UNESCO has provided a range of frameworks and guidance over the last few years to collectively address the gaps. These include the 2019 *Beijing Consensus on AI and Education*, the 2019 *I'd Blush If I Could* publication on AI's gender implications, the 2021 *AI and Education: Guidance for Policy-Makers*, and the 2021 *Recommendation on the Ethics of AI* (UNESCO 2019, 2021a, 2021b, 2023). Together, these documents promote a humanistic approach to AI governance in education, with inclusion, equity, diversity and quality as guiding principles. Similarly, the OECD (2021) emphasised that effective AI integration requires systemic efforts to strengthen assessment, pedagogy and workforce-relevant learning design. More recently, UNESCO IESALC and IBM (2024) have co-developed an AI Competence Framework to help educators and students acquire ethical, context-sensitive AI skills, with particular attention to the Global South (UNESCO IESALC and IBM 2024).

Regionally, the European Union's AI Act classifies educational uses of AI as 'high-risk', requiring transparency, human oversight and accountability measures (European Commission 2024). A recent survey found that 80% of institutions across six European countries—Portugal, Germany, Finland, the Netherlands, Croatia and Denmark—either lacked AI policies or were unsure if they existed (Media and Learning Association 2023). Some of the reported barriers included insufficient resources, institutional resistance and complexity of implementation (Media and Learning Association 2023).

Other countries have also responded to the growing presence of GenAI in higher education. Australia's national regulator, the Tertiary Education Quality and Standards Agency (TEQSA), issued system-wide guidance in 2024 requiring all higher education providers to submit institutional action plans addressing the risks posed by generative AI. Providers were expected to align these plans with national quality standards, conduct course-level risk assessments and integrate AI-related risks into institutional governance processes (Tertiary Education Quality and Standards Agency 2024).

In another study that focuses on East Asian countries, Xie et al. (2024) reviewed GenAI-related policy developments in China, Japan, Mongolia and the United States. China and Mongolia emphasised national sovereignty and safety, while Japan and the United States focused more on improving educational methods and efficiency through a human-centered approach. The researchers also noted that despite shared references to inclusion and equity, few countries had specific policy mechanisms, particularly in addressing infrastructural and digital access gaps (Xie et al. 2024). Moreover, China's policies on GenAI mainly address service providers and users without specifying educational stages, while Japan focuses on higher education and targets university administrators and faculty. In contrast, U.S. policies emphasise K-12 education and address a broader audience, covering various AI technologies beyond generative AI.

2 | Institutional Policies

Although international organisations like UNESCO and the OECD have issued guidance on ethical and equitable AI use in education, these frameworks often address AI broadly, rather than grappling with the unique pedagogical and ethical complexities introduced by generative AI (GenAI). Studies claim that higher education institutions worldwide are increasingly developing policies to regulate their own use in teaching and learning. For example, the University of Hong Kong (HKU) has taken a more comprehensive approach to using GenAI in teaching and learning (University of Hong Kong 2023). In Australia, the University of Sydney has adapted its policies to allow students to use AI tools in open-book assessments (University of Sydney 2023). More recently, after analysing documents among 116 U.S. universities, more than one-third have provided guidelines for classroom teaching (McDonald et al. 2025).

While these examples suggest growing institutional engagement with GenAI policy, a broader look reveals ongoing challenges. Jin et al. (2025) surveyed 40 leading universities across Asia, Europe, North America and Oceania and found that while institutional leaders generally expressed positive attitudes toward the integration of GenAI, formal policies often lacked clarity, coherence or actionable detail. Similarly, Xiao et al. (2023) examined QS ranking top 500 universities; at the time, less than one-third of these universities had some regulations in place; among these universities, more than twice of the universities banned ChatGPT than embraced it. In Canada, Da Mota (2024) reported that there is no existing or proposed legislation that addresses AI use in research institutions, and the draft Artificial Intelligence and Data Act offers no specific guidance for protecting valuable scientific, cultural and historical information in the research institution context. In the absence of national policy, universities such as McMaster University, the University of Montreal and Western University have developed multi-tiered responses: some offer basic classroom guidelines, while others focus on academic integrity, promote AI literacy or establish formal task forces and ethical frameworks. However, a recent report that surveyed Canadian post-secondary institutions asked participants whether their institutions had implemented broad policies on GenAI, and 56% were either unsure or disagreed (Rasky 2024). This gap between formal policy efforts and campus-wide understanding is not unique to Canada—it echoes a broader global trend where the existence

of GenAI guidelines does not necessarily translate to meaningful guidance for day-to-day academic practice.

3 | U.S. Federal and State Policies

Before the GenAI era, there was no specific law for traditional AI in education at the federal level, but agencies like the U.S. Department of Education have issued guidelines on data privacy, fairness and equity that influenced institutional AI policies (U.S. Department of Education 2025). Meanwhile, state policies regarding traditional AI varied widely. For example, California required transparency in AI decision-making, pushing universities to adopt stricter governance (University of California 2025). In states with fewer specific AI regulations, universities developed their own policies, leading to uneven practices across the U.S. (BCLP Report 2025).

With the rise of generative AI and its expansion across higher education beyond traditional STEM fields, federal efforts such as the *AI Leadership to Enable Accountable Deployment Act* have started to establish regulation frameworks for AI, including in educational settings (National Conference of State Legislatures 2023). The National Institute of Standards and Technology (NIST) released an *Artificial Intelligence Risk Management Framework* that includes a profile for GenAI to promote responsible use (National Conference of State Legislatures 2023). Some states like California and Oregon have updated or introduced new laws addressing the use of generative AI in education, with clearer guidelines on academic use (Dusseault and Lee 2023). In contrast, some states such as Illinois and Iowa do not provide state-level guidance, leaving policy decisions to individual institutions at that time (Dusseault and Lee 2023).

4 | Students and Faculty's Perceptions of AI Policies

GenAI adoption among students is widespread and may occur without institutional rules, for example, Coffey (2023) found students reported that they would continue using ChatGPT and other Gen AI tools even if they are banned. At the same time, research on students' perspectives on AI policies remains largely unexplored, leaving a gap in understanding how these policies affect their learning and engagement. Rasky (2024) stated students generally have minimal involvement in shaping Gen AI policies, despite the significant impact these policies have on their educational experience. Other studies have shown that institutional policies regarding Gen AI often use ambiguous language (e.g., 'unauthorized AI assistance'); lack clear guidance and support on when and how Gen AI tools can be appropriately used (Fawns et al. 2024; Hashmi and Bal 2024; Klein 2024; Michel-Villarreal et al. 2023). It is also concerning that Gen AI may pose challenges to academic integrity as students adapt to different policies and regulations across various courses (Mowreader 2024). Tyton Partners (2023) questioned this even further arguing that those policies mostly focused on academic integrity while neglecting issues like bias in large language models and accessibility challenges. All sources point to gaps on policy development and fragmented implementation. Tyton Partners (2023) argue that to address these gaps, institutions

need to refine initial Gen AI policies through collaboration and knowledge sharing (Tyton Partners 2023).

Faculty perceptions of AI policies also remain underexplored; most research has focused on their views regarding AI usage. While faculty generally hold positive attitudes toward AI, concerns primarily centre on academic integrity, critical thinking and potential over-reliance (Xie et al., Under Review; Ghimire et al. 2024). Disciplinary differences also shape faculty perspectives on AI, with STEM instructors typically exhibiting a more positive stance compared to their non-STEM counterparts (Parviz 2024). Despite growing awareness, faculty adoption of AI tools has been slow. A recent report indicates that only 22% of faculty members incorporate AI tools into their instructional practices (Tyton Partners 2023). Luckin and Cukurova (2019) highlighted that many educators lack a sufficient understanding of AI, leaving them vulnerable to ineffective implementation and excessive reliance on technology companies—many of which have limited expertise in teaching and learning. Similarly, inadequate AI training remains a major barrier, with only 10% of surveyed faculty members expressing confidence in using AI tools (Toppo 2024).

5 | Research Problem and Theoretical Framework

While previous studies have highlighted issues such as inconsistent AI policies, unclear guidelines (Mowreader 2024), equity concerns (Xie et al. 2024) and enforcement challenges (Media and Learning Association 2023), many of these discussions have been rooted in traditional AI frameworks and approached from a top-down perspective. They tend to emphasise the role of institutional leaders and administration in crafting and disseminating AI-related policies, often lacking understandings of how these policies are interpreted or applied by those at the ground level, such as faculty and students in higher education. To ensure that both faculty and student perspectives are adequately represented, this study examines how AI policies are understood and enacted in two U.S. higher education institutions through an integrated theoretical framework combining: (1) Policy Enactment theory to analyse institutional policy interpretations and (2) Technology Acceptance models to identify adoption barriers.

5.1 | Policy Enactment Framework

We adopt the Policy Enactment Framework developed by Ball et al. (2012), which emphasises how institutional actors (e.g., faculty, staff, students) reinterpret and reshape top-down policies through local practices, beliefs and constraints. The framework identifies distinct 'policy actor' roles (e.g., translators, resisters) that mediate policy implementation. This perspective aligns with observed disciplinary differences in literature, where STEM faculty historically prioritise research over teaching reforms (Fairweather and Paulson 2008), and generally hold more positive views regarding GenAI than non-STEM faculty (Parviz 2024). In addition, students' widespread use of formally prohibited AI tools (Coffey 2023) shows how policies are interpreted and reshaped by practice. This theory highlights the gap in policy development and implementation and guides the important research significance of student and faculty perspectives.

5.2 | Technology Acceptance Model

To understand individual-level responses to institutional AI policies, this study incorporates elements from the Technology Acceptance Model (TAM) (Davis 1989; Venkatesh and Davis 2000), which posits that perceived usefulness and perceived ease of use influence attitudes and behavioural intentions (Granić and Marangunić 2019). In educational technology research, TAM has been widely extended through additions of external constructs like subjective norm (Park 2009); university support and computer self-efficacy (Al-Mushasha 2013); enjoyment (Chang et al. 2017; Zhang et al. 2008); and student attitudes (Granić and Marangunić 2019). Scholars also noted the importance of providing training on how to effectively use technological resources is recommended to enhance acceptance, while noting the need to consider differences across user groups and cultural contexts in acceptance studies (Granić and Marangunić 2019). Empirical findings from recent literature illustrate how these TAM components apply to GenAI adoption. Faculty adoption remains low (22%) according to training shortages reported by Toppo (2024) likely constraining ease of use. Again, students' continued use of briefly banned AI (Coffey 2023) may suggest that perceived usefulness may override institutional regulations.

While much of the existing GenAI policy research has focused primarily on document analysis, relatively little attention has been paid to the perspectives of faculty and students—those who are responsible for executing these policies in daily practice. By centering their voices, this study intends to bridge the gap between policy development and implementation, examining how institutional AI policies are perceived, negotiated and applied on the ground.

This investigation is guided by these questions,

1. How do faculty perceive the development and effectiveness of AI regulation at their institutions?
2. How do faculty perceive their colleagues' attitudes toward AI at their institution?
3. How do faculty describe institutional support or resources with AI use?
4. What are perceptions of undergraduate students toward AI regulation at their institutions and in their classrooms?
5. How do students' perceptions of AI regulation, as well as their perceived AI-related resources and AI limitations awareness, vary across different demographic groups including gender, year of study and discipline?

6 | Methods

6.1 | Research Design

This study is considered a multiple methods design; the data is part of a larger mixed-methods research project investigating faculty and students' utilisation of AI tools in higher education. During the design phase, while the student survey followed a structured format rooted in TAM constructs, the faculty interviews were more open-ended (follows the survey sections),

allowing participants greater flexibility to reflect on and react to institutional policy guidelines. Data collection took place convergently over a 45-day period in Spring 2024. Although the use of partial data limits the study's ability to fully integrate findings in a way that characterises mixed-methods research, it allows researchers to explore a multi-perspective view of GenAI policy implementation across institutional roles.

Given the distinct participant groups and differing theoretical orientations, the researchers believe this design reflects multiple methods, more specifically a triangulation-based framework (Turner et al. 2015), in which 'different methods are used to capture different elements of the focal phenomenon' (p. 603). In addition, the researchers utilised a parallel analysis approach which each dataset was analysed independently. Nonetheless, the findings were systematically compared in the Discussion to identify thematic convergence, contrast and complementarity. The interpretive integration aligned with a triangulation-based multiple-methods framework.

6.2 | Research Sites

This study took place in two R1 public research universities in the United States; both of the institutions responded rapidly to the emergence of Gen AI during Spring and Summer 2023 and have developed formal AI guidelines for the Fall 2023 semester. Although these two institutions were selected due to researchers' professional connections, these two institutions offer contrasting profiles that enhance the relevance and representativeness. One institution has prominent programmes in the Arts and Health (University A). The other is nationally recognised for STEM majors, especially in engineering and computer science (University B). While we do recognise the limited representativeness of the two research sites in terms of generalising to other institutional types, the inclusion of the two R1 universities provides meaningful insight into how Gen AI policy is experienced within large public research-intensive settings.

6.3 | Qualitative Phase

Interviews have been conducted with seven faculty members to gain insights into institutional-level regulations and their colleagues' perceptions of AI usage. Interview questions include: (1) How do you perceive your university guidance on AI? (2) What about your colleagues' attitude? (3) What support, services or training do you need to better prepare your AI-integrated teaching? (4) What are some sources of education or guidance you have received for incorporating AI in your teaching?

Faculty participants were selected through researcher networks at each institution. The researchers considered two institutions' disciplinary priorities and ensured an even distribution of participants across the two institutions. The participants were purposefully invited using maximum variation sampling to represent a diverse range of disciplines, including both STEM and non-STEM fields; as well as varying levels of experience, from assistant to associate professors (See Table 1). Among the participants, five faculty members reported being very familiar with AI, while two

indicated having some familiarity with AI. Six faculty members reported that they have not implemented AI in classroom teaching; therefore, classroom-level AI policy was not discussed from the faculty perspective in the paper. The researchers acknowledge that it is difficult to fully represent all faculty perspectives with a small sample, participants were asked about their colleagues' attitudes toward Gen AI to represent wider discourse.

Several major codes in the qualitative analysis were aligned with the study's theoretical frameworks. For example, the *Institutional AI Policies and Guidance* code draws directly on Policy Enactment Theory, noting how faculty interpret and adapt to institutional GenAI guidelines. In addition, *Faculty Attitudes Toward AI, Support and AI in Research/Professional Work* reflect key constructs from the Technology Acceptance Model, such as perceived usefulness, ease of use and organisational support. The codebook was further refined through a review of the transcripts (See Table 2 for codebook). Two researchers conducted independent data coding and interpretation. After reviewing the coding of the first transcripts, codes were discussed, and the codebook was updated accordingly. For example, sub-code *AI_training* was added under major code *AI_ethical_consideration* to capture gaps in AI training and development that could lead to inequities in AI literacy. Interrater reliability was calculated, and no differences were found at the major code level.

After coding, we conducted a thematic analysis focused on identifying patterns and variations across participants. We examined how and where faculty perspectives progressed, converged, diverged or were challenged by institutions and also colleagues. While we looked for common themes across interviews, we also paid attention to outlier insights that offered unique perspectives, even if they were not widely echoed.

6.4 | Quantitative Phase

6.4.1 | Data Clean

Responses for international student status and different types of academic purposes (e.g., writing aid) were converted into a binary variable (Yes = 1, No = 0) for analysis. For categorical

variables that include multiple categories and cannot be coded as binary, they were recoded based on the number of respondents. To facilitate analysis while preserving meaningful group distinctions, race was recoded into three categories: Asian, White and Others (including American Indian or Alaska Native, Black/African American, Hispanic/Latinx, Middle Eastern or North African, Native Hawaiian/Other Pacific Islander, Other and Biracial). Similarly, participants' years of study were categorised into lower division (i.e., freshmen and sophomores) and upper division (i.e., juniors and above).

6.4.2 | Participants

Survey flyers were posted in multiple locations across two institutions, and participants were incentivised through a drawing for eight Amazon gift cards. There is a total of 124 undergraduate students who responded to the survey invitations. Table 3 presents the demographic breakdown of survey participants, along with publicly available institutional profiles. Among these student participants, 29.8% identified as male, 55.6% as female, and 14.5% as non-binary or preferred not to answer. In terms of year of study, 35.5% were in the upper division group, while 63.7% were in the lower division group. Regarding majors, 57.3% were enrolled in STEM programs, and 39.5% in non-STEM programs. Participants' racial identities included 29% Asian, 14.5% identifying as 'Other', 37.9% White and 18.5% Multiracial. The total percentage for certain categories does not sum to 100% due to missing data.

When comparing participant demographics to institutional undergraduate data (Table 3), STEM majors were slightly overrepresented compared to University A but aligned with profiles of STEM-focused campuses (University B). Asian students were overrepresented relative to some institutions but consistent with those serving more Asian or international students. Gender diversity was notable, with a higher proportion of non-binary students in our sample than typically reported/found in public reports. While the sample may not be fully representative of all undergraduate students, it reflects diverse U.S. institutional profiles with limited generalisability.

6.4.3 | Data Analysis

To examine the underlying structure of students' perceptions of AI-related factors, an exploratory factor analysis (EFA) was first conducted. Prior to conducting EFA, assumption checks were performed to ensure the suitability of the data. The Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy was 0.66, with individual item values ranging from 0.53 to 0.75. Bartlett's test of sphericity was significant, $\chi^2(231) = 673.86$, $p < 0.001$. These results confirm that the data were appropriate for factor analysis. Based on factor analysis results, items with low factor loadings (below 0.40) were removed.

Figure 1 presents the final factor structure derived from the EFA result. Three latent constructs were identified: Perceived AI Policy (Cronbach's $\alpha = 0.73$), AI Limitations Awareness (Cronbach's $\alpha = 0.60$) and AI Resources (Cronbach's $\alpha = 0.55$).

TABLE 1 | Faculty profile.

Participant pseudonym	Discipline	Familiarity with AI
Steve	Education	Very familiar
Steward	Education	Very familiar
Brooke	Education	Somewhat familiar
Dylan	Political Science	Very familiar
Ning	Physics	Somewhat familiar
Wen	Computer Science	Very familiar
Xiaoming	Electrical Computer Engineering	Very familiar

TABLE 2 | Qualitative codebook.

Major-level code	Sub-level code	Definition	Example
1. Institutional AI Policies and Guidance	1.1 AI Policy Development	Discussions about the existence, development or lack of institutional policies on AI usage in teaching and professional work.	As far as use rules and university guidance, I know that is in development. And there may be some sort of draft versions of guidelines.
	1.2 Policy Clarity and Implementation	Faculty perceptions regarding the clarity, applicability and enforceability of AI-related policies.	So there are guidelines and policies that the university has put out. I think it is just like, they're not enforceable, really.
	1.3 Ethical Considerations in Policy	Concerns about security, academic dishonesty and ethical dilemmas in institutional AI policies.	With employees, there is an emphasis on like security that like we shouldn't be sharing information or uploading things to AI engines.
2. Faculty Attitudes Toward AI	2.1 AI Enthusiasm	Faculty members who actively explore AI's potential in teaching and research.	There are a number of my colleagues who are as excited about it as I am, use it a lot in their work.
	2.2 AI Scepticism	Faculty members who are hesitant to adopt AI, citing concerns about its broader implications.	Some colleagues, I think, are pretty apprehensive of AI. And I think they really see it as bringing about very dramatic change in our social and educational landscape.
3. Support	2.3 AI Opposition	Faculty members who oppose AI usage due to concerns about data control, equity and ethical issues.	Then I have some colleagues who are just very opposed to it for a number of reasons... very valid concerns about who controls the data.
		Faculty training/workshops on AI usage in education.	So I know... have done a very nice series of professional learning on AI. I took part in one of those last year.
4. AI in Research/Professional Work	4.1 AI usage in Research	Use of AI for coding, analysis or research support.	For me, AI tool is not all that different from other types of technology... it is fairly easy to use.
	4.2 AI's Limitations in Research	AI limitations in research or professional work	I tried to get it to like—I fed ChatGPT a description of language coordination... it couldn't do that.
5. AI and Equity Considerations	5.1 AI Training	Faculty concerns about lacking training or AI learning opportunities being optional.	Those opportunities to learn are optional, and they're going to be more attractive to people who already like AI.
	5.2 Inequality	Concerns about AI development being controlled by large corporations and wealth concentration.	Who profits from AI? Which, for the most part right now, are very large companies and very wealthy individuals.
	5.3 Ethical Implications of AI Use	Considerations regarding AI's impact on fairness, bias and decision-making in educational and professional settings.	The data on the internet are clearly biased. And so that bias is inherent in the functioning of AI tools.

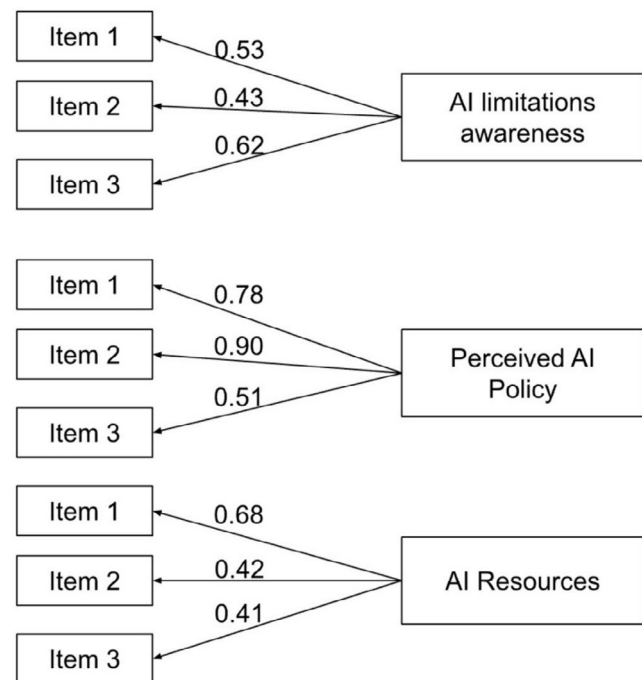
TABLE 3 | Student profile.

Category	Subcategory	Number of students (n)	University A (n = 50)	University B (n = 70)
Gender	Male	29.8% (40)	36.5%	57%
	Female	55.6% (71)	63.5%	43%
	Non-binary/Other	14.5% (18)		
Year of study	Upper division	35.5% (43)		
	Lower division	63.7% (79)		
	Prefer not to answer	(2)		
Discipline	STEM	57.3% (71)	25%	71%
	Non-STEM	39.5% (49)		
	Prefer not to answer	(4)		
Race	Asian	29% (51)	14%	18.9%
	Prefer not to answer/Other race (e.g., Black/African American, American Indian or Alaska Native, etc.)	14.5% (29)	Black or African American: 21% Hispanic or Latino: 11%	Domestic Minority: 12.8%
	White	37.9% (69)	40%	56.4%
	Multiracial	18.5% (23)	—	—

Note: All institutional level demographic percentages were retrieved between 2021 and 2024.

Although the Cronbach's alpha values for AI Limitations Awareness and Perceived AI Resources fall below the commonly cited threshold of 0.70, such values are considered acceptable in research involving newly developed constructs and short scales (Cronbach, 1951; Taber 2018). For example, previous studies have retained scales with α values below 0.60 when measuring constructs with a small number of items (e.g., Van Griethuijsen et al. 2015). In our study, both variables were newly developed, and each was measured using only three items. Given the early-stage development of these variables and the limited number of items, the observed alpha values are within an acceptable range for this exploratory research. While selecting the final items, both theoretical considerations and the study's targeted focus on perceptions of AI policy, limitations and resources were carefully considered to ensure strong conceptual alignment. Thus, each latent variable is represented by three selected items, with standardised factor loadings displayed on the corresponding paths on the figure. For AI Limitations Awareness and AI Resources, item loadings ranged from 0.41 to 0.68, showing moderate contributions to their respective latent factors. For Perceived AI Policy, this study's primary focus, EFA indicated stronger item loadings, ranging from 0.51 to 0.90, suggesting a well-defined factor structure. EFA results suggest a theoretically meaningful and empirically supported structure. Latent variables and corresponding items are displayed in Table 4. Participants rated their agreement with each statement on a five-point Likert scale (1 = Strongly disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly agree).

A Pearson correlation analysis was conducted to examine the relationships among participants' AI-related variables (i.e., perceived AI policy awareness, AI limitations awareness and AI resource usage). Independent samples t-tests were then conducted to examine group differences based on students'

**FIGURE 1** | Factor loadings.

purpose of using AI (e.g., use AI for writing aid or not), academic year (lower-division vs. upper-division students), status (international student or not) and major (STEM vs. non-STEM students) on the identified AI-related variables. Equal variances (homogeneity of variance) are tested by using Levene's test (Levene 1960). When variances were equal, standard t-tests were used. For comparisons where variances were unequal, Welch's t-tests were applied. To further investigate

TABLE 4 | Evaluated variables and corresponding items.

Variables	Items	References
AI Resources	Item 1: I think professors at my college/university support students using AI Item 2: I can get help from others when I have difficulties using AI Item 3: The lecture in my class provides support to help students better use AI in learning (e.g., handouts, website)	Strzelecki 2023
Perceived AI-related policy	Item 1: The classes I took, have policies regarding the use of AI for learning (e.g., classroom activities, assignments and tests). Item 2: The classes I took, have restrictions regarding the use of AI for learning (e.g., classroom activities, assignments and tests). Item 3: My university has clear guidance/policies on using AI with my learning	Jiang et al. 2024
AI limitations awareness	Item 1: AI provides unreliable information or sources for me Item 2: AI responses are biased or exhibit discrimination (e.g., racial or gender) Item 3: I noticed limitations or experienced challenges while using AI	Thi Thuy 2023

group differences based on participants' race, GPA, gender identity, one-way analyses of variance (ANOVA) were conducted to determine whether varying levels of AI usage frequency, GPA or different races were associated with differences in AI resources, perceived AI-related policy and AI limitations awareness. When significant effects were detected, Tukey's HSD post hoc tests were performed to identify specific frequency group differences on variables.

All analyses were conducted by using R studio and assumption checks were performed before each analysis to ensure the validity of the statistical tests. Statistical significance was determined at the $p < 0.05$ level.

7 | Results

7.1 | Faculty Perspective- Qualitative Results

7.1.1 | Institutional AI Policy Development

Except for one faculty member who specialises in policy studies, all the other faculty regardless of their discipline, years of experience admitted that they had not read or incorporated AI policies into their professional work. Faculty members recognised that universities are actively working on establishing and evolving AI-related regulations, as Steve noted, 'As far as use rules and university guidance, I know that is in development. And there may be some sort of draft versions of guidelines', for example languages in syllabus, however, Steve added;

As far as rules from our school or department, we haven't gotten anything specific, although I would imagine that is going to happen. I think as these tools improve, we have to be more specific about the rules of appropriate use and things like that.

Steward also mentioned that AI guidelines exist at the university level, but they are primarily framed within academic dishonesty policies for students and security concerns for employees.

[The university] has AI use guidelines, and they have them both for employees and for students. The students I think it gets wrapped up into the academic dishonesty policy. With employees, there's an emphasis on like security that we shouldn't be sharing information or uploading things to AI engines, because we really don't know what happens with it. Nobody really knows what happens with that information once it's uploaded.

7.1.2 | Institutional AI Policy Implementation

7.1.2.1 | Practicality and Clarity. With existing guidance at institutional level, faculty feel that current guidelines are difficult to enforce, especially regarding practicality and clarity. Ning stated, 'I haven't read any official guidance. I know there were discussions, but I don't think we've received anything practical that actually tells us how to integrate AI appropriately'.

Other faculty have been more specific about the limitations institutions face in regulating AI. Steward stated,

There are guidelines and policies that the university has put out. I think it's just like...they're not enforceable, really. Right? Like, students aren't supposed to use it on assignments—this is in the policy. But again, it's like, how would we ever prove that they did, especially with the tools that exist to detect it being so bad?

Dylan, who also commented on practical difficulty of policy implementation, added,

I tend to think implementation is really hard. It's really messy. If you're not like an AI education person, you may be deleting that email [regarding

AI guidance]. If you're reading it, you may not have enough clarity. So I suspect there are major challenges in the time, capacity, interest, and support for people to implement policies.

Another recurring concern among faculty was the lack of clarity in institutional policies, particularly regarding their applicability to different academic levels. Brooke highlighted this discrepancy, 'I believe that many of those [policies] pertain specifically to undergraduate teaching, which makes sense... I think there is maybe a difference in how you might use AI or think about AI in an intro-level course versus a graduate course'.

7.1.2.2 | Support. In addition to these challenges, faculty expressed the need for more structured institutional support to effectively incorporate AI into their work. While these universities have hosted AI-related workshops, many found them too introductory, theoretical rather than providing concrete practice. Brooke said, 'The most common form of training we've been offered is through professional development workshops. But honestly, they tend to be very general—more like an introduction to AI rather than something I can actually apply to my classes'. Xiaoming echoed, 'So far, what we have are more like discussions. They explain AI broadly, but we still don't have clear, structured training on how to actually use it in teaching or research'.

The other gap faculty highlighted is that while professional development opportunities exist, these workshops are voluntary, thus only attracting those already interested in AI. Steve pointed out, 'Those opportunities to learn are optional, and they're going to be more attractive to people who already like AI... You might have this sort of selection effect where people who are already interested participate, but other faculty do not'. Xiaoming is not among those people. 'Actually, the university sent out emails about some things and also held some workshops, but I didn't attend them. Because I have not used AI because I don't fully understand it'.

Faculty also mentioned that institutional support should be extended to providing advanced access to high-quality AI tools. While free versions of AI tools exist, faculty felt that universities should invest in more advanced AI tools in academic settings. Ning said, 'If the university truly wants us to use AI, they should be willing to invest in it. Right now, we have to rely on free versions, which are limited compared to more specialized AI tools designed for education and research'.

7.1.3 | The Call for Federal-Level AI Policies

Given these institutional challenges, some faculty members advocated for AI regulations at the state or national level, arguing that universities alone lack the authority and resources to regulate AI effectively. Steward emphasised this need,

What honestly needs to happen is, like, probably federal-level policies and guidelines and laws about

restricting what kinds of things AI can do, what kinds of things it says no to, right?

They continued to argue that AI regulation should be directed at developers and corporations rather than being left to individual universities or faculty members.

Because I don't think at the institutional level or the individual faculty level, there's a whole lot that we can do to put the correct safeguards in place...So because then you can actually enforce limits on the AI model makers rather than it being like the university.

7.1.4 | Faculty and Colleagues' Attitudes Toward AI Use

Faculty members mentioned their colleagues' varied attitudes toward AI integration in higher education, which may also reflect institutional uncertainty and inconsistent policy guidance. While some faculty actively embraced AI as a teaching and research tool, others remained sceptical especially regarding ethical concerns, policies and support. As Dylan explained, colleagues' attitude 'ranges from that all the way to like people doing direct instruction AI in their classes...' to 'They wanted the university to block access to things like ChatGPT on the university network'.

Xiaoming's colleagues in Computer and Electrical Engineering are technical experts in AI; they saw it as an assistant rather than a replacement for human intelligence:

I think their attitude toward AI is probably the same as mine because we should be considered as people who understand AI. Most of us more or less know what's going on with it. They definitely see AI as a helpful assistant, but they don't believe it can truly handle many important tasks on its own.

Steve explained this disparity in more detail, some faculty members are seeing AI as a potentially transformative tool. "There are a number of my colleagues who are as excited about it as I am, use it a lot in their work and are also just interested in it. Wanna find different ways that it might be useful or helpful." Others expresses apprehension about AI, "Some colleagues, I think, are pretty apprehensive of AI. And I think they really see it as bringing about very dramatic change in our social and educational landscape that we may not have control over. And I think that is a very valid concern." Finally, A subset of faculty members actively opposes AI adoption in academia, they argue that AI is not simply a neutral tool, but a biased system.

Then I have some colleagues who are just very opposed to it for a number of reasons. I think there are some very valid concerns about who controls the data that are useful in building these AI models where the data come from.

Steve added, ‘There are some very good questions around who profits from AI, which for the most part right now are very large companies and very wealthy individuals’. These colleagues’ opposing attitude are indeed valuable when industry focuses on automation, academic institution is expected to question technology’s broader impact on society, as Steve added,

I would argue that academics are a little more attuned to ethical issues of these tools than perhaps private industry or other companies, where the tool is just sort of embedded in your workflow...We have lots of people who are very interested in ethics and morality and people who study philosophy. And so I think that’s actually a really good thing. I think that’s a healthy perspective to ask these kinds of questions.

7.1.5 | AI With Research

Even though faculty’s AI utilisation is not directly asked in these interview questions, faculty members widely acknowledged AI’s role in enhancing research efficiency, particularly in automating routine tasks such as data analysis, literature reviews and coding assistance. Steve described AI as a valuable tool for streamlining research processes.

For me, AI tools are not all that different from other types of technology. The idea of a chat interface with an AI tool is fairly easy to use. You can upload things, you can insert links and things like that. So I feel pretty well prepared to use the current version of the tools.

Ning explained how ChatGPT helped them in coding,

I asked physicists from other institutions who are numerical experts, they deal with computer programs a lot to solve [problems in] physics. Usually [the code] would take them a few days to write and debug. If you ask GPT 4, it gives you the right answer in a few minutes or a few seconds. So that greatly simplified their lives. I’m not a numerical expert, but I use ChatGPT to debug my own python code. Or [when] I try to install some python package, it gives me all the detailed explanation and troubleshooting with me.

Despite these advantages, faculty also identified limitations, particularly in qualitative research and qualitative data analysis that requires contextual understanding. Brooke gave an example,

I fed ChatGPT a description of language coordination, which requires a lot of context... It couldn’t do that. And I understand because that’s highly sophisticated. And you need a lot of context information to understand what coordination looks like in this particular setting.

Given these challenges, faculty emphasised the importance of AI literacy in research settings. While some institutions offered professional development opportunities, faculty expressed concerns that these training sessions often lacked practical guidance. Steward explained that,

I think that the things that I’ve gone to have been at the level of like, almost hypothetical or it’s either at the level of like 101—like there’s such a thing as a language model—or it’s at this level of like, what are the big picture things that we need to think about? And...I think it would be useful for a lot of people to have a little bit more nuts and bolts [to have] these kind of conversation.

7.1.6 | AI Equity Considerations

While discussing policies, another common theme emerged are concerns about AI’s implications for equity and accessibility in education, particularly regarding disparities in training opportunities. As noted in the previous section (AI Policy Implementation-Support), while some institutions provided AI-related workshops, faculty noted that participation tended to be voluntary, leading to uneven levels of AI literacy among educators. Economic disparities were also discussed (Faculty and Colleagues’ Attitudes toward AI Use)—they argue who benefit from new technology, and also universities’ ability to adopt AI tools especially access to advanced AI models often requires costly subscriptions (AI Policy Implementation-Support). While some faculty members advocated for AI detection tools to prevent academic misconduct, others pushed back against their effectiveness (AI Policy Implementation—Practicality and Clarity). In addition to economic barriers, faculty also raised concerns about biases in AI-generated content and decision-making. Steve pointed out that AI tools are inherently shaped by the data they are trained on, which can introduce significant biases, ‘The data on the internet are clearly biased, and so that bias is inherent in the functioning of AI tools that we have’.

These concerns extended beyond accessibility and fairness to broader discussions on AI’s potential societal impact. At the same time, Wen highlighted that despite ongoing debates about AI’s implications, fears of AI surpassing human intelligence or replacing jobs might be overstated, ‘I remain optimistic about AI’s development, but I don’t think there’s any real need to worry that it will ever reach human intelligence or replace a significant number of jobs’.

7.2 | Students Perspective- Quantitative Results

7.2.1 | Descriptive Statistics

Students generally feel that there are relevant policies (79.8%) and restrictions (77.4%) in place at their institution, but only 16.9% of the participants reported that their class has specific alterations or add-on assignments for students who choose to use AI for learning. Regarding the clarity of these policies and guidance on students’ usage of AI in learning, less than half of the students (48.4%)

consider that they are clear. In addition, a large proportion of participants (81.5%) feel that using AI could potentially lead to being perceived as cheating by their teachers, and many have heard of instances where students were punished for using AI in their learning process, such as exams or assignments.

Table 5 presents the descriptive statistics for the three AI-related variables: perceived AI policy, AI limitations awareness and AI resources. Among these variables, AI policy was the primary focus of this study. It was assessed using three items that measured students' perceptions of class-level and university-level AI-related policies. Specifically, students were asked how much they agreed with the statement on restrictions and policies in their class or university about using AI to assist learning, such as classroom activities, assignments or tests. The results indicate that more than 77% of participants agreed that the classes they attended had policies regarding AI usage, and 78% of the undergraduate students reported having restrictions on using AI for learning. Regarding university-level policy, only 47.5% of participants agreed that their institution has clear guidance or policies regarding AI use in learning. Detailed proportions of participants' responses on each item were included in Table 6. Table 7 shows the correlation matrix among the AI-related variables and reveals that the correlations among the three variables were weak.

7.2.2 | T-Tests

Several independent sample t-tests were conducted to examine whether students' AI-related variables differed by categorical variables such as year of study, race and major. Significant findings are reported below. Welch's t-tests were conducted to examine differences in students' responsible AI usage across

academic years. Lower-division and upper-division students also differed in their perceptions of AI-related policies, 95% CI $[-0.78, -0.08]$, $t(58.95) = -2.47$, $p = 0.017$, with a medium effect size (Hedges's $g = -0.53$). Lower-division students reported significantly higher awareness of AI-related policy compared to upper-division students ($M = 11.4$, $SD = 1.82$ vs. $M = 10.1$, $SD = 3.19$). The comparison of year-of-study groups (lower and upper division) revealed no significant differences in AI limitations awareness or AI-related resources. These findings suggest that students' attitudes toward AI, including its perceived limitations and available resources, remain consistent across different stages of undergraduate study.

In addition to students' years of study, their perceptions of AI policy also differed based on AI usage for academic tasks. Students who chose to use AI as a writing aid reported significantly different perceptions on AI policy than those who did not, $t(112.62) = 2.95$, $p = 0.0038$, 95% CI $[0.39, 1.97]$. Students who used AI for research assistance also scored significantly differently on perceived AI policy compared to those who did not, $t(53.93) = 2.04$, $p = 0.0464$, 95% CI $[0.02, 2.16]$. Both effect sizes for perceived AI policy on AI writing aid usage (Hedges's $g = 0.49$) and AI research assistance usage (Hedges's $g = 0.45$) were small. More specifically, students who did not use AI as a writing aid reported a higher awareness of AI policy compared to those who did ($M = 11.63$, $SD = 1.51$ vs. $M = 10.46$, $SD = 2.88$). Similarly, students who did not use AI for research assistance had higher AI policy awareness compared to those who did ($M = 11.28$, $SD = 2.18$ vs. $M = 10.19$, $SD = 2.91$).

TABLE 5 | Latent variables descriptives.

	Mean	Median	SD	Min	Max
Perceived AI policy	10.95	12	2.46	2	15
AI limitations awareness	7.21	7	2.65	1	12
AI resources	8.75	9	2.42	3	15

TABLE 6 | AI policy descriptives.

Perceived AI policy	Item	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
Class policy	The classes I took, have policies regarding the use of AI for learning	3.2%	4.8%	11.3%	54.0%	25.8%
Class restriction	The classes I took, have restrictions regarding the use of AI for learning	2.4%	6.5%	13.7%	56.5%	21.0%
University policy	My university has clear guidance/policies on using AI with my learning	8.9%	19.4%	23.4%	39.5%	8.1%

Note: $N = 124$.

TABLE 7 | Correlation matrix.

	Perceived AI policy	AI limitations awareness	AI resources
Perceived AI policy	1.00	0.027	0.037
AI limitations awareness	—	1.00	0.140
AI resources	—	—	1.00

Note: $N = 124$.

Other than perceived AI policy, significant group differences were also found among STEM and non-STEM major students' perceived AI-related resources, 95% CI [0.11, 1.82], $t(118) = 2.24$, $p = 0.027$. The small effect size (Hedges's $g = 0.41$) suggests that while STEM students reported significantly higher perceived AI-related resources compared to non-STEM students ($M = 9.25$, $SD = 2.16$ vs. $M = 8.29$, $SD = 2.55$), the difference may not be substantial. However, no significant differences were observed between STEM and non-STEM students in the other two AI-related variables (i.e., AI limitations awareness and perceived AI-related policy). These results suggest that while perceptions of AI-related resources vary by academic discipline, students across different fields of study hold relatively similar views regarding AI's limitations and policy of utilisation.

In addition, two significant differences were found in AI limitation awareness between students who used or did not use AI for writing aids (95% CI [-2.03, -0.15], $t(122) = -2.30$, $p = 0.023$) and exam assistance (95% CI [-2.80, -0.02], $t(122) = -2.01$, $p = 0.046$). The effect size was small for writing aids (Hedges's $g = -0.42$) and medium for the usage of exam assistance (Hedges's $g = -0.54$). Findings show that students who use AI as a writing aid reported slightly greater awareness of AI limitations than those who did not ($M = 7.67$, $SD = 2.55$ vs. $M = 6.58$, $SD = 2.67$). Similarly, students who used AI for exam assistance reported greater awareness of AI limitations than those who did not ($M = 8.44$, $SD = 2.16$ vs. $M = 7.03$, $SD = 2.67$).

7.2.3 | ANOVAs

One-way ANOVAs were conducted to examine whether differences in gender identity, race and academic performance (GPA) were associated with variations in AI-related variables. Results indicated a significant effect of gender on both AI limitations awareness ($F(2, 118) = 12.14$, $p < 0.001$) and perceived AI policy ($F(2, 118) = 3.36$, $p = 0.038$). As suggested by Cohen (2013), the effect sizes of ANOVA were interpreted using the conventional benchmarks for η^2 (small = 0.01, medium = 0.06, large = 0.14). A large effect size ($\eta^2 = 0.17$) indicates that gender accounts for 17% of the variance in AI limitations awareness, but it explained a smaller proportion of the variance in perceived AI policy ($\eta^2 = 0.05$), which is classified as a small to medium effect. Post hoc Tukey's HSD tests revealed that both significant gender differences were primarily driven by the non-binary group. ANOVA results also indicated a significant effect of race on AI limitations awareness ($F(2, 98) = 3.40$, $p = 0.037$). The observed effect size for race differences in AI limitations awareness was medium ($\eta^2 = 0.06$), suggesting that race explained a meaningful proportion of variance in students' awareness of AI limitations. Post hoc Tukey's HSD tests showed that the significant difference was primarily between White and 'Other racial' groups ($p = 0.044$), with the 'Other racial' group perceiving significantly more AI limitations. However, no significant differences were observed between other groups.

However, it is important to note that the non-binary gender group (14.5%) and the 'Other' racial group (18.5%) had relatively small sample sizes, which may limit the generalisability of these findings and reduce statistical power. Therefore, although the differences were statistically significant, the observed effects

should be interpreted with caution, particularly regarding their practical implications.

8 | Discussion

The findings of this study highlight significant gaps in both faculty and undergraduate student familiarity with and implementation of AI policies, especially regarding Gen AI in higher education. While previous research has noted the lack of AI-related guidance, many of these studies were conducted before 2022 (e.g., Zawacki-Richter et al. 2019). With the rise of GenAI and its increased accessibility, this study broadens the scope by targeting a wider population in higher education.

The study's findings align with both Policy Enactment Theory and the Technology Acceptance Model (TAM), which together help explain observed gaps in GenAI policy implementation and stakeholder behaviours across both qualitative and quantitative phases. According to Policy Enactment Theory (Ball et al. 2012), institutional policies are not simply implemented as written, but are interpreted and reshaped by local actors—in this case, faculty and students. The qualitative findings show that faculty viewed institutional-level GenAI guidelines as vague or disconnected from their teaching and research needs, especially at the graduate level. Meanwhile, the quantitative results revealed that students more often recognised GenAI-related policies at the classroom level, suggesting that even unclear, faculty serve as 'translators' who clarify institutional expectations into course-specific rules.

The Technology Acceptance Model (Davis 1989; Venkatesh and Davis 2000) also helps explain adoption behaviours identified in both data phases. Faculty in the interviews frequently cited inadequate training as barriers to ease of use, which reinforces TAM's emphasis on institutional support (Al-Mushasha 2013). More specifically, faculty members expressed two major concerns regarding existing training opportunities: (1) while some institutions offer AI-related workshops, faculty noted that participation is often voluntary, leading to disparities in AI literacy; (2) most trainings are intro-level and may not be effectively reaching or resonating with all instructors.

These faculty perspectives align with student survey results showing that AI-related support was poorly perceived, particularly between STEM and non-STEM disciplines. A previous study found that 900 technology faculty members felt their institutions lacked adequate AI guidance for ethical use (Robert 2024); this study potentially extends this finding across both STEM and non-STEM fields. In addition, students' continued use of AI for writing and research tasks, even when policy clarity was low, reflects TAM's assertion that perceived usefulness can override external constraints (Coffey 2023). Finally, the model's extended constructs, such as subjective norms (Park 2009), were evident in faculty reports of polarised colleagues' attitudes and both faculty and students' concerns regarding equity. These include economic barriers to accessing GenAI tools (faculty) and disparities by academic fields and awareness gaps (students). More specifically, STEM students perceived more AI-related support than non-STEM students and those who are more aware of AI policies are less likely to use generative AI for writing and research. Faculty also noted that existing AI policies primarily emphasise

restrictions rather than providing guidance on appropriate and ethical AI use. This reactive stance may result in framing AI as a potential threat and influence faculty and students' usage. Similarly, Tyton Partners (2023) found that AI policies in education predominantly focus on academic integrity, often neglecting issues such as bias in AI systems and the lack of accessibility support for students with disabilities.

Furthermore, faculty expressed concerns that AI policies tend to be more focused on undergraduate students and provide less guidance for graduate students' learning and professional work. An earlier mixed-methods research suggests that faculty members impose stricter regulations on undergraduate students while adopting a more flexible approach towards graduate-level education (Authors et al., unpublished). This may be related to one of the quantitative findings, which indicated that lower-division students (freshman and sophomore) report significantly higher awareness of AI-related policies compared to upper-division students (junior and senior). Given faculty's varying strict levels of regulation regarding AI use across student levels, freshman and sophomore students who entered college right after the launch of ChatGPT may experience stricter restrictions.

Faculty noted their colleagues' attitudes toward AI varied widely, this indeed reflect broader institutional uncertainty. Some faculty embraced AI as a research and teaching assistant; others remained sceptical, noting ethical concerns and institutional support as barriers to effective AI integration. A subset of faculty members actively opposed AI adoption in academia, arguing that AI is not simply a neutral tool, but 'a biased system shaped by the data it is trained on'. Although these faculty members are actively using AI for research, particularly in tasks like literature reviews, data analysis and coding assistance. They also noticed its limitations in qualitative research that requires contextual understanding. Again, faculty expressed frustration that AI training at their institutions is less practical, they need hands-on guidance needed to support research applications. Finally, additional equity considerations were shared. These faculty pointed to economic disparities in AI access, also noting the access to advanced AI models that require costly subscriptions. Bias in AI-generated content and unreliability of AI detection tools to prevent academic misconduct.

Additional quantitative results revealed that students utilising AI for exam assistance and writing aids have a greater awareness of AI-related limitations. AI usage guidelines and educators frequently emphasise the importance of manual review to ensure the accuracy and appropriateness of AI's writing (e.g., Doyal et al. 2023; Yang et al. 2022). Students who use AI for writing assistance and actively review its output tend to have a higher awareness of AI's limitations, as they may detect issues such as factual inaccuracies during the review process (Ma et al. 2023). Additionally, while using AI for exam assistance, the accuracy of AI's response can directly impact students' scores, making its limitations more apparent. Therefore, encouraging students to use AI with proper guidance while critically assessing and verifying AI-generated content can be an effective way to enhance their awareness of AI's limitations (Lin 2024).

Finally, the ANOVA results reveal significant differences in AI policy perceptions and perceived limitations across gender and racial identity. These findings suggest that certain student groups may experience AI policies and limitations differently, potentially due to varying levels of digital literacy, access to resources or prior exposure to AI tools. However, given the limited sample size, further research is needed to explore the extent of these disparities and their implications for policy-making.

9 | Policy Implications

At the institutional level, clearer and more practical policies are needed; not only rules about academic integrity, but also guidance on appropriate and ethical use (especially with graduate-level teaching and research). Universities also should move beyond voluntary, one-size-fits-all workshops and invest in structured, discipline-specific professional development. Finally, AI-related resources should be equitably distributed across departments, especially for students in non-STEM majors. At the national level, faculty called for clearer national or federal guidance, because individual institutions often lack the capacity or authority to regulate GenAI comprehensively. Coordinated efforts across institutions could also provide models for policy implementation, reducing the burden on individual 'actors'. In addition, these dynamics are likely not unique to U.S. institutions. As GenAI adoption spreads globally, similar implementation gaps and interpretive practices are likely to emerge.

10 | Limitations and Future Research

While this study provides valuable insights into AI policy awareness and perceptions in higher education in U.S. institutions, it has several limitations. First and foremost, the findings are based on data collected during a specific timeframe (spring 2024) at two R1 U.S. institutions; institutional-level AI policies may have continued evolving beyond the study period; the limited sample of institutions may not fully capture the AI policy implementations across the U.S. Secondly, although the ANOVA results reveal significant differences in AI policy awareness and perceived limitations across gender and racial identity, the sample size for certain demographic groups was limited. Future research needs to expand on diverse institutional types and potential sub-groups of students to understand the complexity of AI policy implementation, ex. graduate students. Furthermore, with the rapid evolution of AI technology, the discussion of best practices for AI training programmes and instructional integration is also necessary, particularly regarding how institutions can shift from a restriction-based policy approach to one that promotes responsible and effective AI usage.

Finally, this study does not directly address the formation of such policies at the organisational or field level. Institutional theory (DiMaggio and Powell 1983; Meyer and Rowan 1977) offers important insights into how higher education institutions adopt formal policies not only for functional alignment but also to maintain legitimacy within their broader institutional field—often through symbolic or ceremonial adoption that may be decoupled from daily practices. Building

on this, the institutional logics perspective (Thornton and Ocasio 2008) emphasises how competing societal logics—such as professional, market and bureaucratic—structure organisational attention, decision-making and resource allocation. While we reference these perspectives to help contextualise global policy variation (e.g., the bureaucratic rigour of the EU AI Act vs. market-driven pushes in Australia), our study does not empirically examine how such logics operate within the two U.S. institutions under study. Future research could integrate institutional logics more explicitly to examine how organisational priorities and identity influence the creation and positioning of GenAI policies.

11 | Conclusion

This study highlights the complexities surrounding AI policy implementation in higher education from faculty and students' perspectives in the United States. While students recognise that classroom-level policies are more established, institutional AI governance remains ambiguous. Also, classroom policies tend to be restriction-focused; students report greater awareness is associated with less usage of AI for writing and research. Faculty members also struggle to integrate AI into practice due to unclear policies and insufficient training. They noted that 'voluntary' opportunities may widen AI literacy gaps and limit conversations. Faculty also observed that many workshops are introductory. While many faculty acknowledge AI's effectiveness in research, guidance on AI in professional work is less developed compared to undergraduate teaching. Varying attitudes among colleagues further reflect the lack of clear policies. Faculty also advocate for state or federal-level policies to provide more comprehensive guidance.

Additional quantitative findings reveal disparities in AI policy perceptions across disciplines and demographic groups. STEM students report greater access to AI-related support compared to their non-STEM counterparts, while gender and racial identity also significantly influence AI policy awareness and perceived limitations.

Recognising these policy challenges from faculty and students' perspective, this study advocates for the importance of ongoing policy refinement, institutional training programmes and federal-level guidelines to ensure consistent and equitable AI governance. Future research should also assess the effectiveness and equity of AI-related training initiatives and potentially examine how institutions can transition from restrictive policies to ones that promote responsible and beneficial AI integration.

Author Contributions

Yingying Jiang conceptualised the study, contributed to data collection and analysis, and led the writing on the methods, results and discussion section. Lindai Xie contributed to data collection and analysis, and led the writing on the quantitative methods and results. Xiaoyu Cao conceptualised the study, contributed to qualitative data analysis and writing on the literature review section. All authors contributed to the interpretation of the findings, reviewed drafts of the manuscript and approved the final version for submission.

Ethics Statement

This study was approved by the Institutional Review Board at Virginia Commonwealth University (Protocol # HM20028984).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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